

ESPIRITU SANTO-PEKOA, Vanuatu

WMO: 915540

Lat: 15.501S

Lon: 167.223E

Elev: 184

StdP: 14.60

Time Zone: 11.00 (E11)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	66.6	68.8	59.3	76.0	73.6	61.3	81.7	74.4	19.9	78.1	18.6	77.7	2.9	300	0.323

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	11.1	88.0	79.5	87.2	79.2	86.4	78.8	81.0	85.7	80.6	85.5	80.0	84.8	8.7	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.8	155.6	84.2	79.1	152.0	83.7	78.7	149.9	83.4	45.1	86.4	44.4	85.6	43.9	85.2	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	17.1	15.5	63.6	90.7	2.3	2.9	61.9	92.8	60.6	94.5	59.3	96.1	57.6	98.2		
(5)			60.9	82.2	1.9	1.1	59.5	82.9	58.4	83.6	57.3	84.2	55.9	84.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	78.7	80.5	80.6	80.3	79.8	78.5	77.2	76.6	76.3	77.2	78.1	79.2	80.0
	DBStd	2.42	1.59	1.50	1.33	1.40	2.02	2.16	2.11	2.27	1.98	2.06	2.02	1.68
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10471	946	856	941	895	883	817	823	814	817	872	875	932
	CDD65	4996	481	436	476	445	418	367	358	349	367	407	425	467
	CDH74	42144	4753	4342	4573	4130	3514	2571	2133	1974	2470	3235	3881	4568
	CDH80	8691	1374	1283	1239	973	524	233	141	151	234	480	789	1270
Wind	WSAvg	7.0	5.6	5.5	5.1	4.9	6.9	7.9	8.2	8.6	8.9	8.7	7.5	6.2
Precipitation	PrecAvg	91.1	10.8	10.4	12.9	10.0	7.3	6.4	4.3	4.3	4.2	5.6	7.9	7.1
	PrecMax	132.1	20.9	26.6	34.9	34.5	14.1	19.8	15.4	14.8	12.4	15.2	20.1	13.5
	PrecMin	44.8	0.0	0.0	4.8	3.1	0.6	0.1	0.4	0.6	0.1	0.0	1.6	0.9
	PrecStd	21.2	5.7	5.9	6.3	6.2	4.4	4.7	3.5	3.6	3.3	4.0	4.4	3.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.0	89.2	88.2	87.8	86.1	85.1	84.3	84.5	85.1	86.0	87.3	88.2
		MCWB	79.8	79.6	79.7	80.2	78.8	78.0	76.8	76.9	76.5	77.2	78.8	79.4
	2%	DB	87.7	87.9	87.1	86.4	84.8	83.7	82.7	82.9	83.7	84.6	86.1	87.2
		MCWB	79.5	79.4	79.4	79.3	77.9	77.3	75.9	75.9	76.1	76.3	78.0	79.0
	5%	DB	86.5	86.9	86.3	85.7	84.0	82.6	81.5	81.8	82.4	83.8	85.1	86.2
		MCWB	78.9	79.2	79.1	78.9	77.3	76.5	75.2	75.1	75.3	75.9	77.3	78.5
	10%	DB	85.8	86.0	85.6	84.7	82.8	81.5	80.5	80.5	81.2	82.7	84.1	85.5
		MCWB	78.5	79.0	78.9	78.4	76.5	75.7	74.4	74.1	74.5	75.4	76.7	78.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.8	81.9	81.2	81.4	80.3	80.0	78.8	78.4	79.1	79.7	80.3	81.0
		MCDB	86.9	86.4	85.7	85.7	84.4	82.9	81.9	82.4	81.7	82.9	85.1	86.1
	2%	WB	80.8	81.0	80.6	80.5	79.3	78.8	77.5	77.2	77.8	78.4	79.2	80.2
		MCDB	85.8	85.8	85.5	85.3	83.2	82.2	80.8	81.4	81.0	82.2	84.1	85.5
	5%	WB	80.0	80.3	80.0	79.8	78.7	77.8	76.6	76.4	77.0	77.5	78.7	79.6
		MCDB	84.9	85.3	84.9	84.4	82.5	81.2	80.0	80.4	80.5	81.6	83.6	84.8
	10%	WB	79.3	79.7	79.4	79.1	77.9	77.0	75.7	75.5	76.2	76.7	78.0	78.9
		MCDB	84.1	84.6	84.2	83.6	81.6	80.3	79.2	79.2	79.8	81.0	82.7	84.1
Mean Daily Temperature Range	5% DB	MDBR	10.8	11.1	10.4	9.8	8.8	8.1	7.8	7.8	8.4	9.2	9.7	10.9
		MCDBR	12.3	12.7	11.8	11.4	10.2	9.7	9.6	9.9	10.1	10.7	11.3	12.1
		MCWBR	6.0	6.1	5.7	5.7	5.5	5.2	5.2	5.4	5.4	5.3	5.5	5.9
	5% WB	MCDBR	11.5	11.6	11.2	11.1	9.3	8.8	8.9	9.2	8.8	9.6	9.7	11.5
		MCWBR	6.3	6.2	5.9	6.1	5.4	5.5	5.6	5.5	5.2	5.3	5.3	6.2
Clear-Sky Solar Irradiance	taub	0.403	0.395	0.392	0.387	0.392	0.396	0.400	0.399	0.405	0.410	0.408	0.401	
	taud	2.545	2.576	2.590	2.607	2.568	2.553	2.516	2.515	2.489	2.490	2.514	2.537	
	Ebn at Noon	298	298	293	284	271	264	266	276	285	290	295	299	
	Edh at Noon	35	34	32	30	29	29	30	32	35	36	36	35	
All-Sky Solar Radiation	RadAvg	1938	1845	1699	1492	1299	1172	1239	1432	1670	1806	1876	2008	
	RadStd	186	157	157	130	127	77	72	101	145	164	176	142	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page